Executive Architecture & Operational Specification

Platform User Roles Matrix, Autonomous SEO & AI Discovery Engine, and Enterprise Verification Dossier

CORPORATE IDENTITY	REGISTERED OFFICE	LIVE WEB URL	VERIFICATION STATE
CIN: U01100PB2026PTC063294	**Goindwal Sahib, Tarn Taran, Punjab**	**https://farmsking.in**	**100% Passed (Live VPS & Local)**

1. CORPORATE MANDATE & REGULATORY GOVERNANCE

CORPORATE AUTHORITY

FarmsKing Technologies Private Limited is an institutional Agri-Tech corporation incorporated under the Companies Act, 2013, Ministry of Corporate Affairs (MCA), Government of India. The platform is chartered to deliver artificial intelligence diagnostics, real-time APMC Mandi price discovery, certified agrochemical distribution, satellite vegetative telemetry, and logistics networks in strict alignment with approved statutory e-MoA objects.

100% Pure React Mode

60 FPS WEB

Single cross-platform React Native Web architecture engineered for PC desktop screens with fluid responsiveness across mobile viewports.

Autonomous GEO Engine

INDEXNOW LIVE

Real-time submission of 25 canonical routes to Bing, Yandex, and AI knowledge crawlers (ChatGPT, Perplexity, Claude, Gemini).

ISO Quality Model

ISO 9001/27001

Cryptographic JWT authentication, role segregation, tamper-evident audit logs, and ICAR agricultural protocol compliance.

2. ARCHITECTURAL AUTHENTICATION & TRI-PORTAL SEGREGATION

ACCESS ARCHITECTURE

The platform segregates all platform capabilities into three independent security envelopes to prevent role ambiguity and protect data privacy:

1. Client Portal Flow

Accessible via standard web login. Dedicated to Farmers, Gardeners, and general Agricultural Consumers for crop diagnostics, mandi rates, and retail commerce.

2. Partner Portal Flow

Accessible via dedicated Partner Console. Designed for commercial affiliates, agronomy advisors, input retailers, soil testing labs, and drone fleet operators.

3. Admin Executive Console

Restricted administrative back-office for internal corporate staff. Enforces single-department credentials with mandatory two-factor authentication.

3. IDENTITY COEXISTENCE & MULTI-PROFILE TRANSITION RULES

IDENTITY RULES

Cross-Category Coexistence Rule

A verified mobile number can maintain separate profiles across different categories (e.g., an individual can be a Client, a Partner, and a Staff member). Each category routes to its dedicated interface with segregated authentication tokens and data boundaries.

Intra-Category Role Constraints

Within **Client Category**, Customer combines with either *Farmer OR Gardener*. Within **Partner**, *Advisor combines with Agent*. Within **Farm Staff**, *Supervisor can also hold Labour role*. Within **Corporate Staff**, exactly *one role* is permitted.

4. SYSTEM INFRASTRUCTURE & CLOUD ARCHITECTURE OVERVIEW

STACK OVERVIEW

Frontend Core

React Native Web, Expo SDK, hardware-accelerated 60 FPS transitions.

Backend API

NestJS Enterprise Modular Framework, Node.js 22, Express Gateway.

Cloud Database

Neon PostgreSQL Serverless, Prisma ORM, SSL/TLS encrypted pooling.

Production Cloud

Hostinger Dedicated Linux VPS, Nginx Reverse Proxy, PM2 cluster manager.

5. CATEGORY 1: CLIENT ECOSYSTEM (FARMERS, GARDENERS, CONSUMERS)

CATEGORY 1: CLIENTS

The Client Category represents the primary demand-side user base. Engineered with 100% pure English copy and easy wording, this interface enables rural cultivators and urban growers to access sophisticated agricultural services intuitively without technical barriers.

CLIENT ROLE	OPERATIONAL RESPONSIBILITIES	PROVIDED PLATFORM FEATURES	REQUIRED PROFILE & KYC DATA
Farmer COMMERCIAL CULTIVATOR	Manages commercial crop cultivation cycles, monitors field plant health, discovers APMC mandi prices, and orders certified fertilizers/seeds.	• **AI Crop Doctor:** Instant leaf disease detection with precision spray dosages. • **Live Mandi Rates:** Real-time APMC mandi prices and 7-day commodity price trends. • **Satellite Telemetry:** NDVI vegetative index and hyper-local weather alerts. • **Dosage Calculator:** Chemical/fertilizer tank mixing calculations per acre. • **Farmer Khata:** Digital ledger, passbook, and voucher settlement.	• Mobile Number (OTP Verified) • Full Name & Father's Name • Village, Tehsil, District, State, PIN • Land Holding Size (Acres/Killas) • Primary Crops Grown (Wheat/Paddy) • Irrigation Source & Soil Classification
Gardener URBAN HORTICULTURE	Oversees terrace rooftop plantations, kitchen garden plots, indoor ornamental plants, domestic lawns, and urban plant nurseries.	• **Micro-Dosage Guide:** Small-batch organic plant nourishment schedules. • **Pest Scanner:** Household insect and fungi identification engine. • **Potting Mix Telemetry:** Soil aeration and watering frequency advisories. • **Gardening Shop:** Specialized seeds, planters, potting soil, and hand tools.	• Mobile Number (OTP Verified) • Full Name & Residential City/Town • Postal PIN Code • Garden Type (Rooftop / Balcony / Lawn) • Primary Plant Categories Managed
Customer / Buyer AGRI CONSUMER	General consumers and retail buyers purchasing fresh farm produce, verified bio-inputs, seeds, and gardening tools.	• **Agri-Commerce Marketplace:** Farm-direct produce & products. • **Real-Time Order Tracking:** Live shipment tracking from dispatch to doorstep. • **Digital Invoicing:** GST-compliant printable sales bills and A4 slips. • **Flexible Payments:** Razorpay UPI, Net Banking, and Cash-on-Delivery.	• Mobile Number (OTP Verified) • Full Name & Delivery Address • Postal PIN Code & Landmark • Email Address (Optional for e-Invoices)

6. CLIENT ROLE SEGREGATION & MUTUALLY EXCLUSIVE BOUNDARIES

ROLE LOGIC

Farmer + Customer
Permitted. A Farmer retains full buying capabilities to order certified inputs from shop.

Gardener + Customer
Permitted. A Gardener accesses urban plant tools and buys potting supplies in shop.

Supervisor + Labour
Permitted. A Farm Supervisor can also hold the Labour Worker role under their employer farmer.

Farmer vs Gardener
Enforced. An account cannot hold both Farmer and Gardener roles simultaneously.

7. CLIENT ONBOARDING & VERIFICATION LIFECYCLE

ONBOARDING

1. Mobile OTP Auth
Farmer inputs mobile number, receives 6-digit cryptographic OTP, validating carrier identity.

2. Geo-Location Lock
Captures district and Tehsil coordinates to map nearest APMC mandi and regional weather station.

3. Agronomy Baseline
Farmer selects active seasonal crops (e.g. Wheat, Basmati Rice) and landholding acreage.

4. AI Feature Access
Instant activation of AI Leaf Scanner, Dosage Calculator, and Mandi Rate telemetry cards.

8. CATEGORY 2: BUSINESS & SERVICE PARTNER FRAMEWORK

CATEGORY 2: PARTNERS

Partners form the operational backbone of FarmsKing in rural districts. They provide certified agronomy consultations, agrochemical input distribution, professional soil lab testing, drone precision spraying, and commodity trade aggregation.

PARTNER ROLE	COMMERCIAL & FIELD MANDATE	PROVIDED PLATFORM FEATURES	REQUIRED KYC & LICENSING DATA
Agronomy Advisor CROP SPECIALIST	Reviews farmer disease scan inquiries, conducts tele-consultations, and issues certified treatment regimens.	• Diagnostic Consultation Queue • Digital Prescription Authoring Tool • Advisory Fee Wallet & Instant Settlement • Historical Farmer Treatment Logs	• B.Sc / M.Sc Agriculture Degree • Professional Agronomy License • PAN Card & Aadhaar Number • Bank Account Details (Cancelled Cheque)
Franchise / Retailer AGRI-INPUT STORE	Stocks certified fertilizers, seeds, and agrochemicals; fulfills local farmer orders and provides POS billing.	• Live Inventory Management & Stock Sync • Point-of-Sale (POS) & Voucher Printing • Trade Commission Ledger & Instant Payout • Low-Stock & Expiry Telemetry	• Fertilizer / Seed / Pesticide Retail License • GSTIN Registration Certificate • Physical Store Address & GPS Coordinates • Business PAN & Current Bank Account
Soil Testing Lab SOIL LAB OPERATOR	Performs chemical spectrophotometry on soil/water samples and uploads verified soil health report cards.	• Sample Barcoding & Tracking System • N-P-K, pH & Micronutrient Entry Grid • Automated Soil Health Card Generator • Testing Fee Disbursal Wallet	• NABL Accreditation or State Lab Permit • Lab Equipment Calibration Certificate • Authorized Chemist Signatory ID • GSTIN & Commercial Bank Mandate
Logistics Fleet TRANSPORT FLEET	Transports bulk farm harvest from farm gates to mandis, and delivers wholesale supplies to retail stores.	• Route Manifest & Load Dispatch Console • Electronic Proof of Delivery (e-POD) • Distance & Trip Mileage Payouts • Live Fleet GPS Telemetry	• Commercial Vehicle RC & Fitness Certificate • Driver Commercial Driving License (DL) • Vehicle Comprehensive Insurance • Aadhaar & Bank Settlement Account
Drone Spray Pilot AERIAL APPLICATOR	Executes precision aerial spraying of micronutrients and approved crop protection formulations over farmland.	• Flight Booking & Dispatch Scheduler • Acreage Log & Battery Telemetry • Spray Verification Certificate Generator • Pilot Pay-Per-Acre Settlement	• DGCA Remote Pilot Certificate (RPC) • Drone UIN Registration Document • Aviation Third-Party Liability Insurance • Pilot Aadhaar & PAN Card
Buyer / Arhtiya / FPO COMMODITY TRADER	Procures farm produce directly from registered cultivators with transparent digital weighment and instant payout.	• Live Produce Lot Bidding Console • Mandi Cess & Digital Settlement Slips • Escrow Account Disbursal System • Supply Chain Provenance Log	• APMC Mandi Trader License • FPO Registration / Corporate PAN • GSTIN Certificate & Trade Name • Verified Current Bank Account

9. PARTNER DUAL-CREDENTIAL POLICY & FINANCIAL SETTLEMENT

AFFILIATE POLICY

Advisor + Franchise Agent Dual Role

An individual possessing agricultural qualifications can operate as an **Agronomy Advisor while simultaneously serving as an Input Booking Agent**. This enables advisors to recommend protocols and seamlessly assist farmers with input orders.

Automated Escrow Settlement

Platform commission settlement operates through integrated banking webhooks. Payouts for advisory fees, retail margins, and drone spraying clear directly into verified commercial bank accounts upon digital sign-off.

10. CATEGORY 3: CORPORATE STAFF & PLATFORM ADMINISTRATORS

CATEGORY 3: STAFF

Corporate staff accounts are restricted exclusively to direct employees of FarmsKing Technologies Private Limited. In strict compliance with ISO/IEC 27001 Information Security standards, each staff member is assigned exactly one departmental role, audited at every invocation.

CORPORATE ROLE	DEPARTMENTAL MANDATE	ADMIN CONSOLE PERMISSIONS	CORPORATE CREDENTIAL REQUIREMENTS
Super Admin EXECUTIVE HEAD	Platform Governance, Cloud Infrastructure, Master Financial Approvals, and Statutory Compliance.	• Full unrestricted platform access • Master feature flag toggles • System audit log & security overrides • Database backup & restore controls	• Board Resolution Authorization • Official Corporate Email (@farmsking.in) • Hardware FIDO2 2FA Security Key • Corporate NDA & Confidentiality Deed
Agronomy Operations AGRONOMY DIVISION	Audits diagnostic accuracy of AI engines, oversees verified advisor quality, and reviews treatment efficacy.	• Advisory quality assurance desk • Disease dataset verification & labeling • Advisor turnaround time monitoring • Banned chemical override controls	• M.Sc / Ph.D in Agronomy/Plant Pathology • Corporate Employment Appointment Letter • Corporate ID & Role Token
Mandi Desk Analyst MARKET INTELLIGENCE	Verifies daily APMC market price feeds, detects pricing anomalies, and publishes commodity trend reports.	• Mandi rate review & publishing console • Anomaly detection & flag management • Historical price trend analysis tools • Commodity volume arrival tracker	• Degree in Agri-Economics / Statistics • Verified Staff Appointment Letter • Corporate Access Credentials
Support Specialist CUSTOMER EXPERIENCE	Provides live dispute resolution for farmers, processes delivery complaints, and manages refund vouchers.	• Unified customer support ticket desk • Voice/call log escalation interface • Refund & replacement voucher authorization • Farmer account recovery console	• Background KYC & Police Verification • Customer Experience Certification • Corporate CRM Identity Token
Supply Chain Manager SUPPLY OPERATIONS	Supervises central inventory warehouses, vendor delivery SLAs, and regional distribution hub logistics.	• Central inventory & warehouse console • Vendor purchase order approvals • Logistics transit milestone audit • Damaged goods & return settlements	• Supply Chain Management Degree/Diploma • Verified Corporate Employment Deed • Departmental Access Key
AI Data Scientist RESEARCH & AI	Trains and benchmarks computer vision disease models, tunes satellite NDVI algorithms, and optimizes latency.	• Model inference telemetry monitor • Confusion matrix & validation review • Training pipeline dispatch console • Edge-model weight deployment logs	• M.S. / B.Tech in Computer Science / AI • Security Background Check Clearance • Cloud AI Cluster Access Credentials

11. STRICT STAFF INVARIANT: SINGLE ROLE MANDATE & SECURITY RBAC

SECURITY POLICY

Zero Role Blending Invariant

Unlike the Partner or Client ecosystems, a Corporate Staff account is cryptographically restricted to **strictly one active role**. A Mandi Analyst cannot access Support tickets or Super Admin settings.

Tamper-Evident Audit Logging

Every query, status update, price adjustment, and refund executed within the Admin Console is permanently stored in the audit ledger with operator ID, IP address, and timestamp.

12. AUTONOMOUS REAL-TIME SEO & GENERATIVE ENGINE OPTIMIZATION (GEO)

SEARCH & AI VISIBILITY

To ensure maximum web search rank and full conversational AI discoverability, FarmsKing incorporates an autonomous multi-engine indexing pipeline:

IndexNow Real-Time Protocol — HTTP 202 ACCEPTED

Using a validated 32-character hexadecimal key (`7c23f4c12cd8c5dcddf6d149b8339b9f`), the system submits all 25 canonical routes to Bing, Yandex, and IndexNow central hubs automatically every 12 hours.

Generative Engine Profiles (GEO) — SCHEMA.ORG @GRAPH

Specialized discovery profiles (`llms.txt`, `llms-full.txt`, `ai-knowledge.json`) index 17 platform capabilities with direct deep-links, sub-second inference metrics, and verified corporate metadata.

SEO & AI FEATURE	IMPLEMENTATION ARCHITECTURE	LIVE PRODUCTION VERIFICATION
IndexNow Protocol Key	32-character hexadecimal key deployed at root with strict newline suppression.	`https://farmsking.in/7c23f4c12cd8c5dcddf6d149b8339b9f.txt`
Recurring Submitter Service	NestJS `SeoModule` submitting 25 routes every 12 hours + startup dispatch.	`GET /api/v1/seo/status` (Active, 25 URLs)
AI Discovery Profile	Structured feature catalog with markdown deep-links for conversational LLMs.	`https://farmsking.in/llms.txt`
Technical AI Dossier	Full architectural specification with ICAR agronomy seals and latency metrics.	`https://farmsking.in/llms-full.txt`
Structured Knowledge Graph	Schema.org JSON-LD graph with Corporation, WebSite, and 17 Service catalogs.	`https://farmsking.in/ai-knowledge.json`
Robots.txt AI Directives	Explicit permissions for GPTBot, PerplexityBot, ClaudeBot, and Googlebot.	`https://farmsking.in/robots.txt`

13. ENTERPRISE VERIFICATION & SYSTEM HEALTH AUDIT

PRODUCTION STATUS

All production microservices, public web portals, administrative consoles, and local development clusters have been validated with zero faults:

Live Production Backend — HTTP 200 OK

Endpoint: https://farmsking.in/api/v1/health
Latency: 42ms • DB Connected
Security: HSTS, CSP, Helmet Active

Live Web Application — HTTP 200 OK

URL: https://farmsking.in/
Framework: React Native Web (Expo)
Cache: No-store, instant revalidate

Local Dev Stack — PORTS 3000 / 8081 / 8082

API: Port 3000 (Active)
Frontend: Port 8081 (Active)
Admin Portal: Port 8082 (Active)

FARMSKING TECHNOLOGIES PRIVATE LIMITED

CIN: U01100PB2026PTC063294 • Regd. Office: VPO Goindwal Sahib, Tehsil Khadur Sahib, Tarn Taran, Punjab - 143422

Institutional Standard: ISO 9001:2015 (Quality Management) • ISO/IEC 27001:2022 (Information Security)

AUTHORIZED SIGNATORY
Board of Directors • Technology Division